

carbon and water.

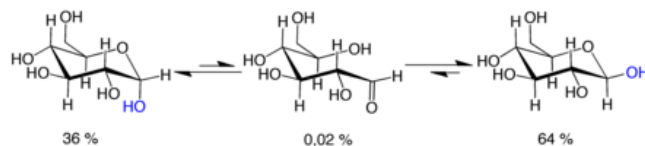
Rotational isomers

Each glucose isomer is subject to rotational isomerism. Within the cyclic form of glucose, rotation may occur around the O6-C6-C5-O5 torsion angle, termed the ω -angle, to form three staggered rotamer conformations called *gauche-gauche* (gg), *gauche-trans* (gt) and *trans-gauche* (tg).^[25] There is a tendency for the ω -angle to adopt a *gauche* conformation, a tendency that is attributed to the *gauche* effect.

Mutarotation

Mutarotation consists of a temporary reversal of the ring-forming reaction, resulting in the open-chain form, followed by a reforming of the ring. The ring closure step may use a different -OH group than the one recreated by the opening step (thus switching between pyranose and furanose forms), or the new hemiacetal group created on C-1 may have the same or opposite handedness as the original one (thus switching between the α and β forms). Thus, though the open-chain form is barely detectable in solution, it is an essential component of the equilibrium.

The open-chain form is thermodynamically unstable, and it spontaneously isomerizes to the cyclic forms. (Although the ring closure reaction could in theory create four- or three-atom rings, these would be highly strained, and are not observed in practice.) In solutions at room temperature, the four cyclic isomers interconvert over a time scale of hours, in a process called mutarotation.^[26] Starting from any proportions, the mixture converges to a stable ratio of α : β 36:64. The ratio would be α : β 11:89 if it were not for the influence of the anomeric effect.^[27] Mutarotation is considerably slower at temperatures close to 0 °C (32 °F).



Mutarotation: D-glucose molecules exist as cyclic hemiacetals that are epimeric (= diastereomeric) to each other. The epimeric ratio α : β is 36:64. In the α -D-glucopyranose (left), the blue-labeled hydroxy group is in the axial position at the anomeric center, whereas in the β -D-glucopyranose (right) the blue-labeled hydroxy group is in equatorial position at the anomeric center.

Optical activity

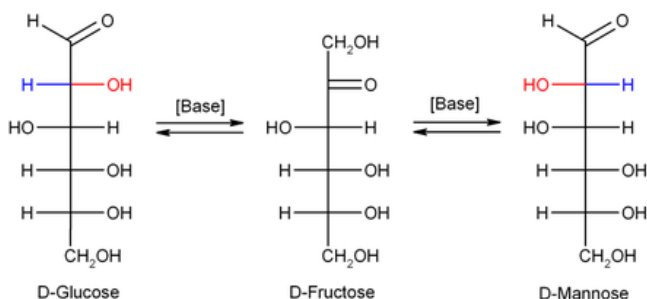
Whether in water or in the solid form, D-(+)-glucose is dextrorotatory, meaning it will rotate the direction of polarized light clockwise as seen looking toward the light source. The effect is due to the chirality of the molecules, and indeed the mirror-image isomer, L-(−)-glucose, is levorotatory (rotates polarized light counterclockwise) by the same amount. The strength of the effect is different for each of the five tautomers.

Note that the D- prefix does not refer directly to the optical properties of the compound. It indicates that the C-5 chiral center has the same handedness as that of D-glyceraldehyde (which was so labeled because it is dextrorotatory). The fact that D-glucose is dextrorotatory is a combined effect of its four chiral centers, not just of C-5; and indeed some of the other D-aldohexoses are levorotatory.

The conversion between the two anomers can be observed in a polarimeter, since pure α -D-glucose has a specific rotation angle of $+112.2^\circ \cdot \text{ml} \cdot \text{dm}^{-1} \cdot \text{g}^{-1}$, pure β -D-glucose of $+17.5^\circ \cdot \text{ml} \cdot \text{dm}^{-1} \cdot \text{g}^{-1}$.^[28] When equilibrium has been reached after a certain time due to mutarotation, the angle of rotation is $+52.7^\circ \cdot \text{ml} \cdot \text{dm}^{-1} \cdot \text{g}^{-1}$.^[28] By adding acid or base, this transformation is much accelerated. The equilibration takes place via the open-chain aldehyde form.

Isomerisation

In dilute sodium hydroxide or other dilute bases, the monosaccharides mannose, glucose and fructose interconvert (via a Lobry de Bruyn–Alberda–van Ekenstein transformation), so that a balance between these isomers is formed. This reaction proceeds via an enediol:



Rocket Fuel

Dextrose is commonly used in homemade rockets built by amateur rocketeers. The dextrose is commonly mixed with a solid Oxidizer such as Potassium Nitrate To make "Rocket Candy" or the weaker KNDX propellant.

Biochemical properties

Metabolism of common monosaccharides and some biochemical reactions of glucose